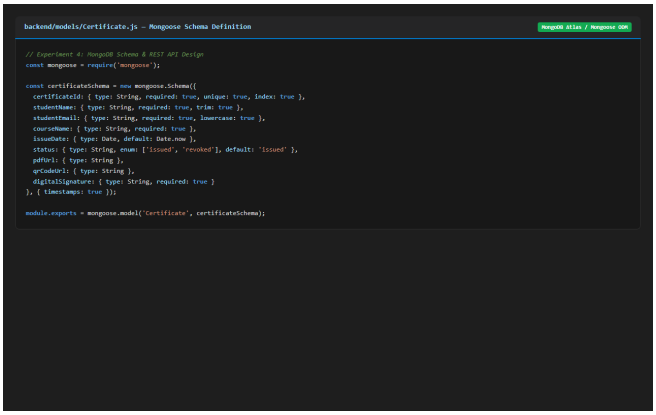
Experiment 4: REST API Design with MongoDB & Mongoose Integration

Aim: Build schema-backed REST API endpoints connected to MongoDB Atlas cloud database with indexing and validation.

Tech Stack / Concepts: MongoDB Atlas, Mongoose ODM, B-Tree Indexes, Timestamps, SHA-256 Signatures

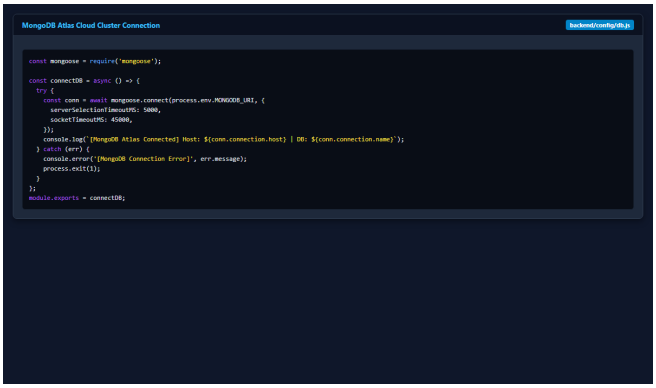
Repository: github.com/JaideepGowda2006/Certified | Student: Jaideep Gowda

Screen 1: Certificate Mongoose Schema & Uniqueness Indexes



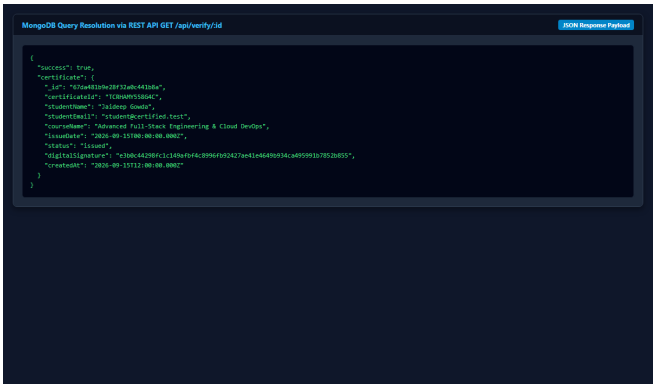
Schema definition enforcing unique certificateId, enum status, and hashes.

Screen 2: MongoDB Atlas Cloud Connection Module



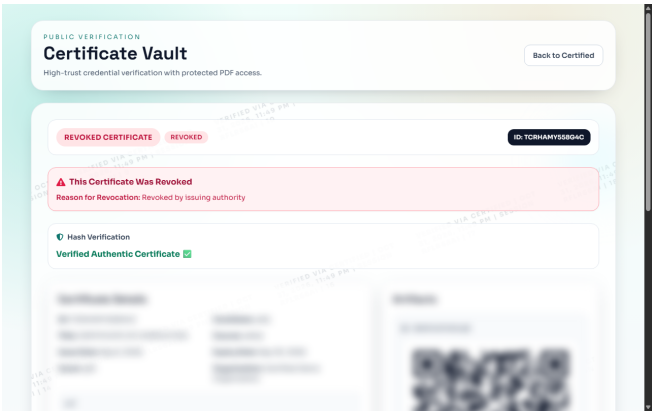
Mongoose connection pool configuration with connection timeout handling.

Screen 3: Cloud Database Query JSON Resolution



Live GET /api/verify/:id returning indexed certificate document from Atlas.

Screen 4: Live Verified Certificate Rendered in Client



Client-side verification view pulling validated attributes directly from MongoDB.